Trade and the Origins of the Territorial State

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September 9, 2026

What determines the emergence of territorial states?

Which economic and social forces lead to the emergence of states?
What *shapes* the geographic extent of the state?

$\rightarrow$ Fundamental questions of political economy

Difficult to study empirically:

- Emergence of states happened a long time ago $\leftarrow$ scarcity of data
- Geographic extent of the state is an equilibrium outcome of inter-state competition $\leftarrow$ boundaries confound economic forces with military competition

This paper: study state formation in 10th century Eastern Europe

Why? Three reasons:

1. **Formation of new territorial states out of tribal chiefdoms**
   - Piast Poland, Volga Bulgaria, later Kyivan Rus
2. **New territorial states are not immediately adjacent to existing territorial states**
   - $\rightarrow$ realized state borders are informative about (relative) payoffs to territorial expansion, and the underlying economic and political forces
3. **Rich archaeological data** on commerce and on state borders of the first Polish state
   - Strongholds (dendrochronologically dated) informative about state boundaries
   - Coin finds informative about trade and other networks

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Roadmap:

1. Linear regressions indicate that changes in market access and trade *caused* formation of territorial states (Poland ∼940, Volga Bulgaria ∼940)
2. Model of trade and spatial extent of the state
   - Coin data to estimate geography of trade flows (cf. Boehm & Chaney, 2026)
   - Boundary of early Polish state allows identification of economic forces

▸ Background

Main findings

1. Market access $\rightarrow$ state formation
   - Changing market access to Islamic silver increased the likelihood of state formation in Poland and Volga Bulgaria, around 940
2. Silver flows *shaped* the state
   - Coin flows constitute taxable traffic along rivers
   - The Piast state in Poland expanded to reach the rivers and tax silver flows
   - Borders reveal the relative returns to taxation and costs of territorial control
3. The form of revenue matters
   - River tolls: about 27% of net land revenue plus toll revenue
   - Predictable silver revenues paid soldiers, enabling further taxation, plunder, and territorial consolidation and expansion
4. Origins of Coercive Capacity among Eastern European states (Tilly '92)
   - Demand for slaves + supply of silver + geography $\Rightarrow$ spatially concentrated silver flows
   - Formation of territorial and coercive states to tax silver flows, and use coercion to get more slaves

Literature: key papers on state formation and state boundaries

- Tilly (1990), Olson (1993), Mayshar-Moav-Pascali (2022, JPE), dal Bo-Hernandez-Lagos-Mazzuca (2022, APSR), Allen-Bertazzini-Heldring (2023, AER), Heldring (2025, ARE)
- **Sanchez de la Sierra (2020, JPE)**: States as stationary bandits
  Key difference: quantification of forces from through trade data and data on border
- **Flueckiger, Larch, Ludwig, Pascali (AER forthcoming)**: Emergence of bronze age city states on trade route bottlenecks
  Key difference: here: territorial states, not city states; borders help w/ identification
- **Fernandez-Villaverde, Koyama, Lin, Sng (QJE, 2023)**: geographic determinants of state borders
  Key difference: causal evidence; equilibrium model allows counterfactuals
- **Allen (WP 2022)**: quant. model of state borders through inter-state competition
  Key difference: here: not inter-state comp. but everything else

Literature: History, Archaeology, Numismatics

Key difference: econometric identification of trade's effect on state formation; estimation of fiscal payoffs from trade flows and the state border.

- **Silver flows and the slave trade:** Noonan (1992); Jankowiak (2020)
  Changing routes and rhythms of exchange with the Islamic world
- **Silver, tribute, and Piast state formation:** Adamczyk (2014, 2020)
  Long-distance trade and silver redistribution as foundations of political power
- **Interpreting coin finds:** Brather (2007); Kilger (2008); Jankowiak (2018)
  Hoard composition and silver fragmentation as evidence of exchange
- **Territory and coercion:** Kara (2020); Biermann (2021); Izdebski et al. (2025)
  Strongholds and territorial control; slavery; social and ecological change

Political actors in Eastern Europe, 10th century

Khazar Khaganate:
6th century – 969 AD

Political actors in Eastern Europe, 10th century

Rus tribes:

Ladoga ~ 750, Novgorod ~ 850,
conquest of Kyiv ~ 865;
Kyivan Rus (870s onwards)

"The Rus raid the *ṣaqāliba*, sailing in their ships until they come upon them, take them captive, and sell them in Khazaria and in Bulgar. They have no cultivated fields and they live by pillaging the land of the *ṣaqāliba*. [...] They have no dwellings, villages, or cultivated fields. They earn their living by trading in sable, grey squirrel, and other furs. They sell them for silver coins which they set in belts and wear round their waists. [...] They treat their slaves well and dress them suitably, because for them they are an article of trade." (Ibn Rustah, Kitāb al-Aʿlāq al-nafīsa, 903–913)

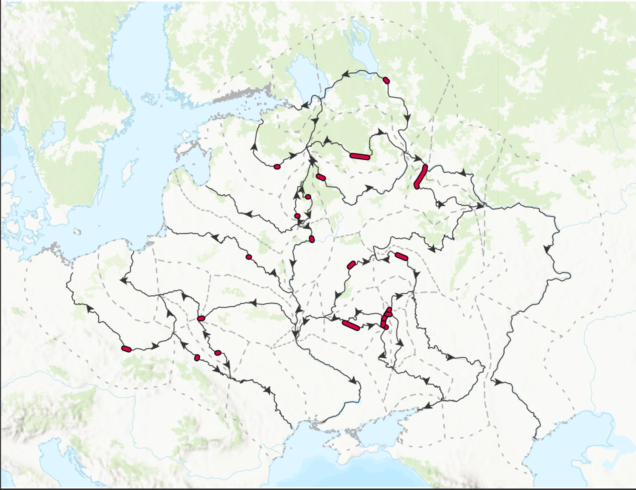
Political actors in Eastern Europe, 10th century

Poland (940s)

Political actors in Eastern Europe, 10th century

Volga Bulgaria (8th/9th c – 1236 AD)

Trade routes in Eastern Europe, 9th–10th century



Slaves, furs, and forest products exchanged for Arab silver

- Overland routes in red
- Rivers in black; arrows show flow direction
- Dashed lines partition space by nearest trade route (up to 200 km)

Sources: Леонтьев and Носов, 2012; Tentiuc, 2020; Channon and Hudson, 1995.

Around 940 AD: Formation of Polish state under Piast dynasty (Mieszko I.)

Ibn Yaqub writes, around 965:

> *"Regarding the land of Mieszko, this is the largest of the Slavs' lands. It is abundant with food – meat, honey and crops. He collects taxes in the form of [al–mathaqil al–marqktija] (silver, by weight). and they are used to pay the monthly salaries of his men [warriors], each of whom receives a fixed number. He has three thousand armoured men and they are divided into military units. One hundred of them is equal to a thousand other [warriors]. He gives them clothing, horses, weapons and everything else they need." [transl., after Kowalski, 2022]*

Hence:

1. Taxation of trade flows on rivers (main taxable source of silver; Adamczyk 2014, Hardt, 2018)
2. Silver used to pay warriors
3. Archaeological evidence shows that Poland was heavily and actively involved in the slave trade as well (Adamczyk, 2014, Roach, 2020, Biermann, 2021)

The setting: Eastern Europe in the 8th-10th century

Extremely data-scarce setting:

1. **Literary sources:**
   - Chronicles: Frankish monks (Bavarian Geographer, ca. 830-870), Byzantine chronicles (Constantine VII's *De Administrando Imperio* $\sim$950)
   - Arab travel reports (Ibn Yaqub $\sim$960s, Ibn Fadlan 922), Khazar letters (King Joseph's letter $\sim$950s)
   - Later chronicles: Rus *Primary Chronicle*: 12th century, Gallus Anonymus: 12th century.
   - Only "official" document is *Dagome Iudex* letter ($\sim$991)
2. **Archaeological sources:**
   - Coins
   - Metal tools, weapons, jewellery (settlements), objects of exchange (glass beads)
   - Pottery
   - Strongholds (dated w. tree-rings)
   - Palnyology (agriculture/deforestation)

Data

$1{,}255$ coin hoards with $193{,}711$ coins, digitized from the numismatic & archaeological literature

- Hoard *tpq* between 725 and 1040 AD
- From present-day Poland, Ukraine, and Moldova (west) to the Volga basin (east)

Coins & hoards show:

- *Where* and *when* a coin was minted
- *Where* and *when* it was buried in the form of a hoard (*terminus post quem*)

Structural econometrics (later) add hoards from Gotland and Bornholm.

▸ Map ▸ Coin statistics ▸ Hoard statistics

Example of a hoard description

nº	Династия.	ПРАВИТЕЛЬ.	ГОРОД.	Год.	Ссылки и особые признаки.	Целые монеты.	Обломки.	Диам.	Вес.
1	Омейяды.	....	?	126	....	—	1		0,83
2	Аббасиды.	Мансур	Мединат-ас-Селам.	156	Т. nº 853, надломан........	1	—	26,0	2,73
3	»	Рашид........	Ифрикия.	[170]	Т. nº 1113........	—	1		0,71
4	»	Рашид	Мединат-Зерендж.	[186—9]	С именем سيف بن الطاراى. Т. nnº 1401, 1419, 1441, 2815.	—	1		0,56
5	»	Рашид........	ал-Мухаммедия.	?	....	—	1		1,02
6	»	Рашид........	Ма'дин-аш-Шаш.	190	Т. nº 1478........	—	1		1,63

Bykov (1925): a hoard found in Novgorod in 1903. Columns report dynasty, ruler, mint, mint year (*anno hegirae*), notes, whole coins, fragments, diameter, and weight.

All coins are Islamic dirhams; hoard *tpq*: 341 AH (952–953 AD).

What do coin hoards tell us?

1. Presence and quantity of finds: coarse measures of access to silver
   - But q'ty also depends on deposition, discovery, and documentation
2. Composition by origin and age: more informative about circulation
   - Coins valued by silver weight → fungible across origins and vintages
   - More young coins → larger recent inflows relative to the existing stock

Long-distance coin flows are largely the result of commerce (Noonan, 1992); Short-distance flows of coins can be due to other reasons as well (plunder, gift, tribute, ransom,...).

I'll use the term "trade" in a broad sense, encompassing these other motives.

▸ Temporal distributions

Reduced-form: does market access/trade *cause* state formation?

Challenge: Trade and state formation may be jointly determined

- Productivity could drive both
- States may facilitate trade $\rightarrow$ reverse causality

Strategy:

1. OLS regressions of state presence on trade indicators from coins
2. IV regressions using shifts in market access:
   - Rus–Khazar conflict: disrupts established routes through Khazaria
   - Silver supply shifts: production shifts from the Middle East to Central Asia

Regression specifications

Region $\times$ 25-year period:

$$\text{StatePresent}_{it} = \beta\, \text{Coins}_{it} + \alpha_i + \alpha_t + \varepsilon_{it}$$

Hoard level:

$$\text{StatePresent}_h = \gamma\, \text{YoungShare}_h + \psi(\text{Size}_h) + \alpha_{i(h)} + \alpha_{t(h)} + \varepsilon_h$$

Coin level:

$$\text{StatePresent}_{h(c)} = \delta \log(\text{Age at deposit})_c + \alpha_{i(h(c))} + \alpha_{t(h(c))} + \varepsilon_c$$

YoungShare: fraction younger than 20 or 40 years; ψ: cubic in log hoard size.

StatePresent: Location contains a territorial state (Poland or Volga Bulgaria after 940)

Region: Area of closest proximity to a given river segment (up to 200km)

OLS: correlation between state formation and coin inflows

	Dep. var.: StatePresent_{it}		Dep. var.: StatePresent_{h}		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0171** (0.0035)				
log Number of Coins		0.0151^{+} (0.0080)			
Fraction of coins $<$ 20 y.o.			0.0980** (0.026)		
Fraction of coins $<$ 40 y.o.				0.135** (0.026)	
Log coin age at deposit					-0.0116** (0.0021)
Hoard Size Controls			Yes	Yes	
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Mint FE					
Unit of Observation	Region $\times$ Time	Region $\times$ Time	Hoard	Hoard	Coin
R^2	0.579	0.391	0.809	0.811	0.904
Observations	442	442	1020	1020	126452

$^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

SEs clustered by region in (1)–(2), by coin group in (5).

More coins and younger coins are associated with state presence.

Instrumental variables: IV #1

Rus–Khazar conflict: disruption of access to Arab traders

$$Z_{it}^{\text{conflict}} = \log\left(\min_{a \in \{\text{Black Sea, Caspian Sea, Bolgar}\}} \text{TravelTime}_{ia,t} \right)$$

- Conflict dated to 925 in the empirical implementation
- From 925: travel speeds through Khazar territory are *halved*
- Baseline speeds: 8 km/h downstream; 1 km/h upstream and at portages

  ▸ Experimental archaeology

- Regions relying on disrupted routes lose relative market access

Identifying assumption: conditional on fixed effects, this exposure potentially affects state formation *only* through market access.

IV results: trade disruption

	Dep. var.: StatePresent_{it}		Dep. var.: StatePresent_{h}		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0331**				
	(0.011)				
log Number of Coins		0.101*			
		(0.042)			
Fraction of coins $<$ 20 y.o.			2.898*		
			(1.14)		
Fraction of coins $<$ 40 y.o.				2.827**	
				(1.04)	
Log coin age at deposit					-0.443**
					(0.096)
Hoard Size Controls			Yes	Yes	
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Mint FE					
Unit of Observation	Region $\times$ Time	Region $\times$ Time	Hoard	Hoard	Coin
First stage w. id. robust F	3.851	6.277	6.477	7.310	18.32
R^2	0.107	-1.804	-11.48	-9.982	-9.923
Observations	429	429	916	916	105837

$^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Trade indicator instrumented by log shortest travel time to Arab traders. SEs clustered by region in (1)–(2), by coin group in (5).

First stages are weak in some specifications.

Instrumental variables: IV #2

Silver supply shifts: Abbasid to Samanid coin production

- From the late 9th century, silver inflows shift toward Central Asia
- Regions differ in proximity to the changing sources of coin supply

$$\text{SilverMarketAccess}_{it} = \sum_j \frac{d_{ij}^{-1}}{\sum_{j'} d_{ij'}^{-1}} s_{jt}$$

- d_{ij}: great-circle distance to the centroid of polity j's mints
- s_{jt}: polity j's share of coins minted in period t and found in the data

Identifying assumption: external silver-supply shifts potentially affect state formation *only* through market access.

▸ Mint maps

IV results: silver market access

	Dep. var.: StatePresent$_{it}$		Dep. var.: StatePresent$_{h}$		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0175**				
	(0.0017)				
log Number of Coins		0.0763**			
		(0.026)			
Fraction of coins $<$ 20 y.o.			4.123*		
			(1.65)		
Fraction of coins $<$ 40 y.o.				4.010**	
				(0.93)	
Log coin age at deposit					-0.515**
					(0.15)
Hoard Size Controls			Yes	Yes	
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Unit of Observation	Region $\times$ Time	Region $\times$ Time	Hoard	Hoard	Coin
First stage w. id. robust F	4.052	7.754	11.99	13.62	57.48
R^2	0.374	-0.876	-23.89	-20.77	-13.61
Observations	429	429	916	916	105837

$^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Exposure-robust SEs from equivalent shock-level regressions (Borusyak, Hull, and Jaravel, 2022).

Positive effects for coin inflows; negative effect for coin age.

Interpreting the reduced-form evidence

- Both instruments support a causal role for access to silver
- Could state formation have triggered the conflict?
  - Historical accounts emphasize Byzantine involvement, not Piast or Bulgar expansion
- Could conflict directly increase demand for a state?
  - Poland remote from the conflict; harder to rule out for Volga Bulgaria
  - Silver-supply IV provides a second source of variation

None of this says that there is only *one* determinant of territorial state formation.

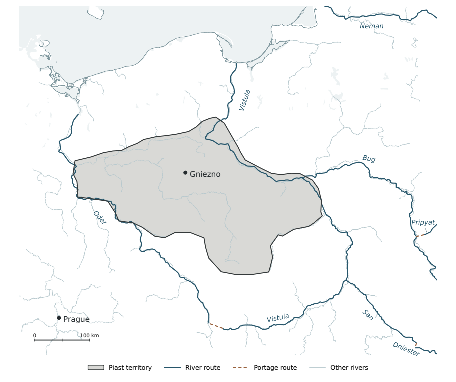
▸ Exposure controls   ▸ Including Rus territories

Part II: Silver, market access, and the shape of the early Piast state in Poland

To better understand the mechanisms that led to the rise of territorial states, we'll focus on one case:

The emergence of the first Polish state under the Piast dynasty and Duke Mieszko I.

First Polish state under the Piast dynasty



Poland, ca. 960. Source: Kara (2020).

▸ Original

1. Territory reaches the Vistula and Oder
   Taxable trade; defensible frontiers
2. Limited expansion toward the Baltic Sea
   Exposure to Viking raids
3. Gniezno lies between the main river corridors
   Cost of collecting revenue at a distance
4. Uneven expansion in the south
   Potential competition with Bohemia

Model components

1. Estimate geography of inter-regional coin flows
    - How large was silver "traffic" along the rivers?
    - Coins of different origins and vintages trace flows
    - Flows can be interpreted as the result of trade (Boehm & Chaney 2026), plunder, or other motives
    - Estimate coin stocks and "trade" shares from hoard counts and composition
    - Assign flows to shortest paths → tax base along each river

2. Model of territorial choice within Poland
    - Given tax base along rivers, ruler chooses territory to maximize revenues net of costs
    - Land taxation and river tolls, collection costs, political competition, frontier defense
    - Fit the observed border → relative importance of these forces

Inter-regional model: household budget constraint

▸ Running out of time...

Households in region i enter period t with a stock of coins $S_i(t)$:

$$S_i(t+1) = \underbrace{(1-\lambda)S_i(t)}_{\text{Surviving coins}} + \underbrace{M_i(t)}_{\text{New minting}} + \underbrace{Y_i(t)}_{\text{Income}} - \underbrace{X_i(t)}_{\text{Expenditure}}$$

- λ: fraction of inherited coins lost, melted, or converted into objects
- $M_i(t) \geq 0$: exogenous *mint mass*, i.e. new coins issued in i at t; zero outside minting regions
- Consumption expenditure $X_i(t)$ needs to be financed with coins ($\rightarrow$ outflow of coins)
- Income $Y_i(t)$ received at the end of the period ($\rightarrow$ inflow of coins)

Assume no saving out of available balances, $s_i(t) = 0$: then

$$X_i(t) = (1-\lambda)S_i(t) + M_i(t) \quad \Rightarrow \quad S_i(t+1) = Y_i(t)$$

Accounting for coin types

A coin type is a *mint location* $\times$ *mint period*

$S_{mi}(\tau, t)$: inherited coins in i at t, minted in m at $\tau < t$.

Fungibility: agents blind to vintages $(m, \tau) \Rightarrow$ each coin type is "spent" in the same proportions. The *flow* of coins of each type is proportional to the *stock*.

$$S_{mi}(\tau, t+1) = \sum_n \left[(1-\lambda)S_{mn}(\tau, t) + M_n(t)\right] \pi_{ni}(t)$$

where

$$\pi_{ni}(t) = \frac{X_{ni}(t)}{X_n(t)} : \quad n\text{'s expenditure share on goods from } i\text{; coins move } n \to i.$$

Data-generating process

1. **Hoard counts discipline the scale of coin stocks**

$$D_i(t) \sim \mathrm{Poisson}(\psi S_i(t))$$

- Assumption: common formation/discovery rate ψ across regions and periods

2. **Hoard composition disciplines coin-type shares**

$$q_h \sim \mathrm{Dirichlet}(\kappa H_i(t)), \qquad S_h \mid q_h \sim \mathrm{Multinomial}(N_h, q_h)$$

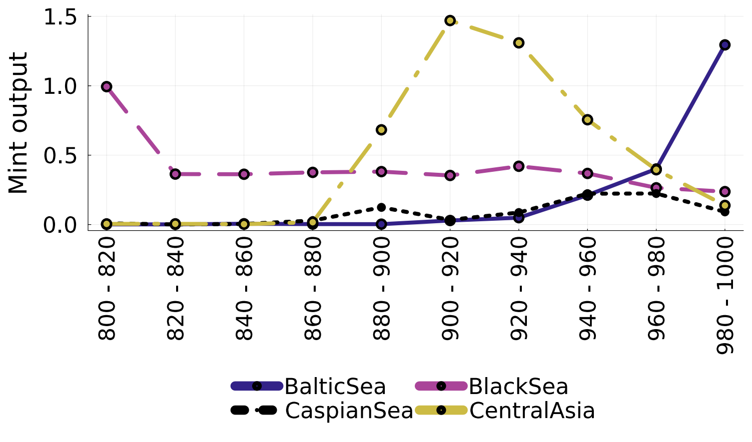
- $H_i(t)$: vector of population type shares $S_{mi}(\tau, t)/S_i(t)$; N_h: number of coins in hoard h
- Finite κ allows coins within a hoard to be more similar than independent draws

Estimate jointly by maximum likelihood. $\hat{\kappa} = 4.99$. Evidence

Implementation of the trade model

1. **Geography and time**
   - 13 interior regions and 4 peripheral minting regions
   - Baltic Sea, Central Asia, Black Sea, Caspian Sea
   - 20-year periods; coin types defined by mint region and mint period
2. **Estimation**
   - Period-specific mint masses and trade shares; overdispersion parameter
   - Nonzero trade: interior/Baltic goods exporters selling to minting regions
   - Hoards from 1000–1039 help identify earlier flows; report estimates before 1000
3. **River traffic**
   - Recover expenditures and trade flows from coin stocks and shares
   - Assign each flow to its shortest path through the river network

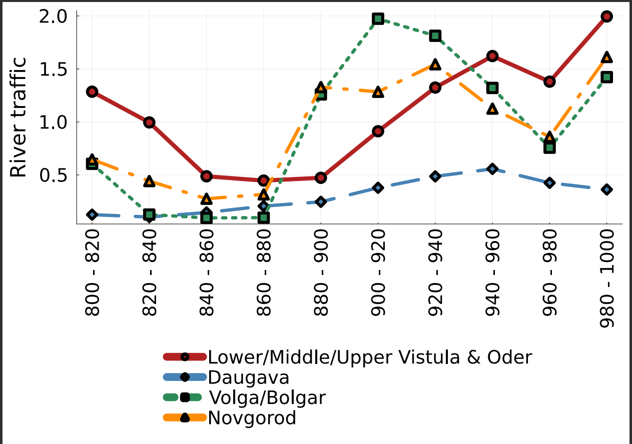
Estimated sources of silver: Mint volumes $M_i(t)$



Central Asian inflows rise; Baltic inflows become sizable late in the 10th century.

Estimated mint masses reaching the study area, not total production at the mints.

Estimated river traffic



Polish river traffic rises from 900–920 and exceeds the other plotted regions from 940.

▸ Polish estimates ▸ Alternative specification

Territorial choice block: revenues

Poland is a graph of locations V and directed edges E. Choose which cells to claim, $x_i \in \{0,1\}$; Gniezno (c) is always claimed, $x_c = 1$.

Ruler's problem: max revenue minus costs $\max_{x \in \{0,1\}^{|V|} : x_c = 1} \Pi(x)$ (Alesina-Spolaore '97).

Production tax revenue, net of collection costs

A claimed cell yields 1 before transport; value falls with return travel time $\mathfrak{t}_{ci}$ to the capital, at rate σ.

$$+\sum_{i \in V} (1 - \sigma \mathfrak{t}_{ci}) x_i$$

River toll revenue

Claiming a river cell yields $\tau \Xi_{r(i)}$: toll payoff per unit of traffic τ times estimated traffic $\Xi_{r(i)}$.

$$+\sum_{i \in V} \tau \Xi_{r(i)} x_i$$

River traffic is taken as given; $\Xi_{r(i)} = 0$ away from taxable rivers.

Territorial choice block: costs

Viking raids
Expected loss on claimed land: ω at the Baltic coast, declining with travel time $\mathfrak{t}_i^{\text{sea}}$ from the sea.

$$-\sum_{i \in V} \omega e^{-\mathfrak{t}_i^{\text{sea}}/\kappa} x_i$$

Competition with Bohemia
Cost is ρ at Prague and declines with return travel time $\mathfrak{t}_i^{\text{prague}}$; may reflect a behavioral wedge rather than a resource cost.

$$-\sum_{i \in V} \rho e^{-\mathfrak{t}_i^{\text{prague}}/\kappa} x_i$$

Frontier defense
Cost λ per edge from claimed to unclaimed land; discounted by $e^{-\beta_R}$ on eligible sides of held river cells.

$$-\sum_{(i,j) \in E} \lambda e^{-\beta_R R_{ij}} x_i (1 - x_j)$$

Estimation and computation

Choose parameters to minimize disagreement with the digitized Piast territory $\bar{x}$:

$$\hat{\theta} \in \arg\min_{\theta} \sum_{i} \mathbf{1}\{\bar{x}_i \neq x_i^*(\theta)\}$$

- Equivalent to MLE under independent cell misclassification with a common probability below $1/2$
- 73-by-62 grid; 10 km cells; Gniezno fixed as capital
  Combinatorial optimization problem: 2^{4526} choices, but:
- Costly frontier defense ($\lambda > 0$) makes the ruler's problem *graph-cut representable* and solvable by a minimum cut in almost-linear time

▸ Min-cut = max-flow

What identifies the model parameters?

1. **Overall extent from Gniezno** $\rightarrow$ spatial decay σ
2. **Reaching rivers with different traffic volumes** $\rightarrow$ toll payoff τ
3. **Northern reach toward the Baltic** $\rightarrow$ Viking cost ω, decay κ
4. **Southern reach toward Bohemia** $\rightarrow$ competition cost ρ, decay κ
5. **Compactness and alignment with rivers** $\rightarrow$ frontier cost λ, river discount β_R

The border data recovers *relative* payoffs in the objective function.

▸ Counterfactual borders

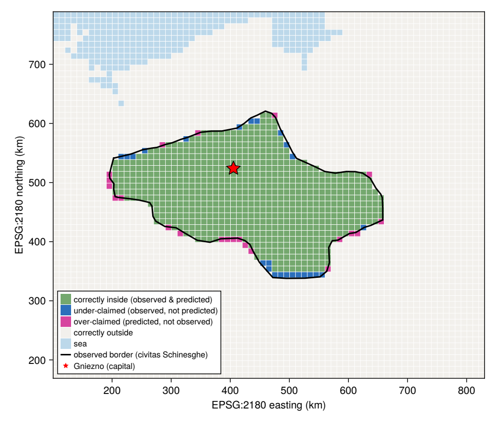
Parameter estimates and model fit

1. **Production taxes**: collection breaks even at 15 travel days from Gniezno

2. **River tolls**: a high-traffic river cell is worth 6.1 ordinary land cells

3. **Viking threat**: coastal cost is 2.1 cell tax revenues; halves every 3.2 travel days

4. **Frontier defense**: 1.7 cell tax revenues per land edge; 39% of this on eligible sides of held river cells



▸ Full parameter table

The composition of the ruler's payoff

Description	Mathematical object	Value	% of (⋆)	resource or shadow cost?	in-kind or monetary?
Net revenues from land tax (⋆)	$\sum_{i\in V}(1-\sigma \mathfrak{t}_{ci})x_i$	328.9	100.0	resource	revenues in-kind costs monetary
Revenues from river tolls	$\sum_{i\in V}\tau_r \Xi_{r(i)} x_i$	124.5	37.9	resource	monetary
Foreign political competition cost	$-\sum_{i\in V}\left(\omega e^{-\mathfrak{t}_i^{\mathrm{sea}}/\kappa_V} + \rho e^{-\mathfrak{t}_i^{\mathrm{prague}}/\kappa_P}\right)x_i$	-189.3	-57.6	either possible	either possible
Frontier defense	$-\sum_{(i,j)\in E}\lambda e^{-\beta_R R_{ij}} x_i(1-x_j)$	-153.0	-46.5	resource	monetary
Sum	Π	111.14	33.8		

- Land revenues are the largest component, predominantly in kind
- River tolls are monetary: 125 units, or 27% of land plus toll revenues
- Political competition may capture behavioral wedges as well as resource costs

Linking river tolls to macro estimates of silver inflows

The border identifies *relative* fiscal returns, not the level of silver receipts.

Additional assumption: on average, 10% of river traffic is paid as a silver toll

- Documented for Khazars and Volga Bulgars (although not known for Poland)

Combining the trade and territorial models, averaged over 940, 960, and 980:

- River tolls: about 0.16 units; total coin inflows: about 0.7 units

$\rightarrow$ Roughly one quarter of silver from tolls, three quarters from other sources (most likely exports of slaves and forest products)

Origins of the territorial state: what did we learn?

1. **Market access change made river locations valuable and triggered state formation**
   - Route disruptions and shifts in Islamic coin production redirected silver
   - Market access change increased return to controlling rivers
2. **Market access mattered for both emergence and territorial extent**
   - Regional evidence: improved silver access increased state formation
   - Territorial model: taxable traffic helped draw the Piasts toward the rivers
3. **Access to money/silver was not enough for territorial state formation**
   - Example of Rus (who had silver but developed a territorial state only much later) shows that money/silver was not enough
   - Most likely explanation is that silver from tribute/plunder was not sufficiently predictable to allow switch to paid retainers
4. **A small silver revenue stream could increase coercion & a larger tax base**
   - Land taxes were largely in kind; the military retinue was paid in silver
   - Paid warriors collected taxes, defended border, obtained captives and export goods
   - Long-term impact of market access shocks

Trade financed coercion $\rightarrow$ coercion generated further tax and export revenues.

Origins of Coercive Capacity among Eastern European States (Tilly ’92)

Tilly (1992): Eastern European states were first coercion-intensive; Western European states were first “capital”-intensive, and build coercive capacity (military) by taxing “capital”.

Tilly (1992) takes existence of states as given (990 CE – 1990 CE).

This paper shows that the coercive nature of Eastern European states is intimately linked to their formation:

- Demand for slaves + availability of silver, concentrated river geography
- Emergence of coercive territorial states to tax silver flows
- Coercion was also used to obtain more slaves (and sell them for silver)

Still, commerce was what initially gave rise to coercion.

THANK YOU!

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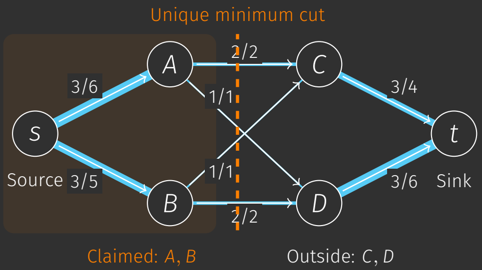
BACKUP SLIDES

Territorial choice as a min-cut problem

Let v_i be land and toll revenue net of collection and threat costs, and $d_{ij} = \lambda e^{-\beta_R R_{ij}}$ the cost of a frontier edge:

$$\Pi(x) = \sum_i v_i x_i - \sum_{(i,j)\in E} d_{ij} x_i (1 - x_j).$$



Width = capacity; label = flow/capacity.

Crossing pipes do not meet. Illustrative values.

Encode the ruler's payoffs

$v_i > 0$: pipe $s \to i$ of capacity v_i.
$v_i < 0$: pipe $i \to t$ of capacity $-v_i$.
$i \to j$: capacity d_{ij} for frontier defense.

The cut is the border.

Source = Capital;
Sink = a point far outside

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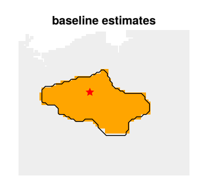
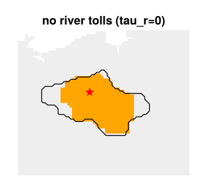
Counterfactual borders

Baseline

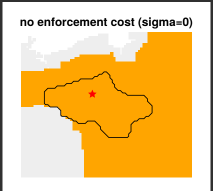


No river tolls ($\tau = 0$)

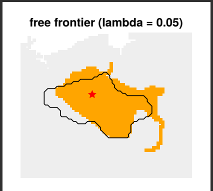


No Viking threat ($\omega = 0$)

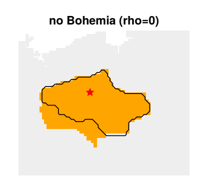
No spatial decay ($\sigma = 0$)



Low defense cost ($\lambda = 0.05$)



No Bohemia ($\rho = 0$)



Orange: predicted territory; black: observed border; star: Gniezno.

Other parameters and river traffic held fixed.

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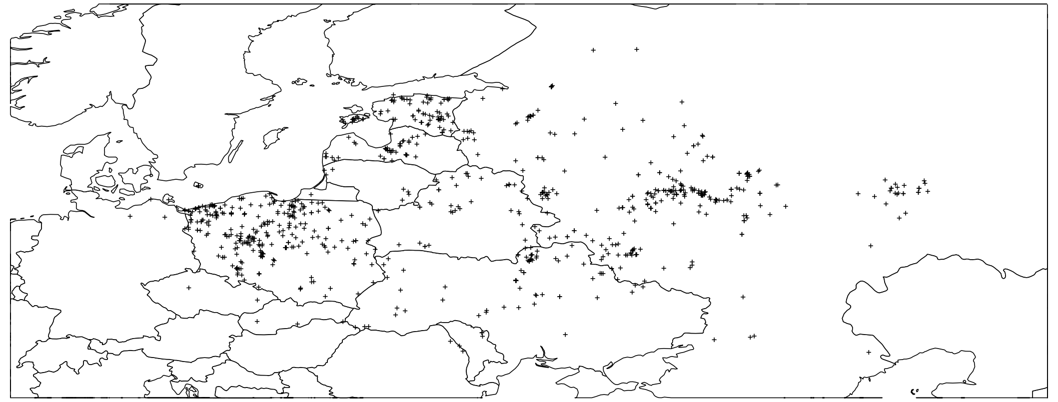
Geographic distribution of coin hoards



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Coin summary statistics

	count	mean	sd	min	p10	p50	p90	max
Has mint date interval	138,558	0.56	0.50	0	0	1	1	1
Has mint location	138,558	0.42	0.49	0	0	0	1	1
Has mint location and date interval	138,558	0.39	0.49	0	0	0	1	1
Mint date interval, years	76,905	18.21	37.16	0	0	1	63	210
Mint date interval, start year	77,268	881.89	75.84	54	771	905	968	1,025
Mint date interval, end year	77,240	899.95	70.80	57	796	912	983	1,200
Age at tpq	76,905	38.24	40.04	-100	4	29	85	929
Has material	138,558	0.57	0.50	0	0	1	1	1
Coin is gold	138,558	0.00	0.00	0	0	0	0	1
Coin is silver	138,558	0.57	0.50	0	0	1	1	1
Coin is copper/bronze	138,558	0.00	0.01	0	0	0	0	1
Has denomination	138,558	0.87	0.34	0	0	1	1	1
Has some empire/dynasty information	138,558	0.99	0.08	0	1	1	1	1
Geodesic distance mint to hoard, km	130,521	36.05	36.43	0	6	21	79	227

Tabulated sample: 138,558 coins. Smaller sample than the full dataset.

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Hoard summary statistics

	count	mean	sd	min	p10	p50	p90	max
Hoard tpq	962	921.27	67.29	743	811	940	996	1,000
Number of coins in hoard	962	144.03	796.05	1	1	8	275	20,966
Fraction of coins with mint date interval	962	0.83	0.27	0	0	1	1	1
Fraction of coins with mint location	962	0.53	0.40	0	0	1	1	1
Fraction of coins with mint date interval and mint location	962	0.51	0.40	0	0	1	1	1
Average mint date interval	961	17.22	31.25	0	1	4	52	200
Average age of coins at tpq	961	22.07	33.31	-100	-0	16	60	464
Fraction of coins with material	962	0.66	0.47	0	0	1	1	1
Fraction of coins that are gold	962	0.00	0.04	0	0	0	0	1
Fraction of coins that are silver	962	0.65	0.47	0	0	1	1	1
Fraction of coins that are bronze	962	0.01	0.10	0	0	0	0	1
Fraction of coins with denomination	962	0.83	0.35	0	0	1	1	1
Fraction of coins with empire/dynasty information	962	1.00	0.01	1	1	1	1	1
Average distance of coins from mint, km	953	44.19	39.32	0	5	32	92	227

Tabulated sample: 962 hoards, *tpq* 743–1000. The full dataset covers 725–1040.

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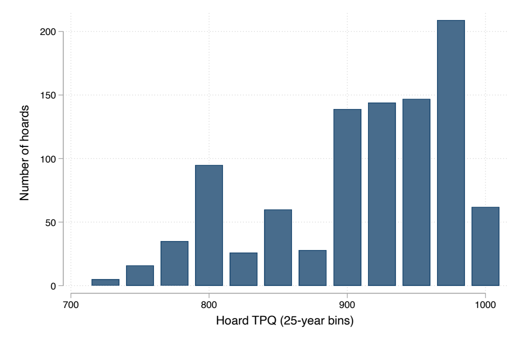
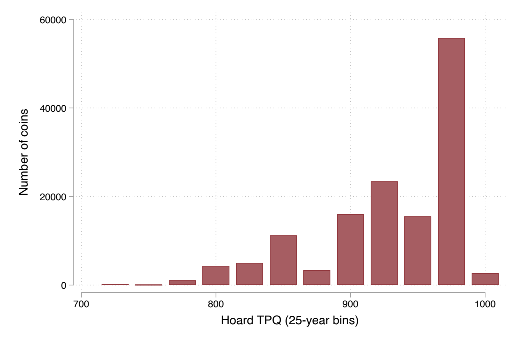
Temporal distribution of hoards and coins

Hoards by *tpq*



Coins by hoard *tpq*



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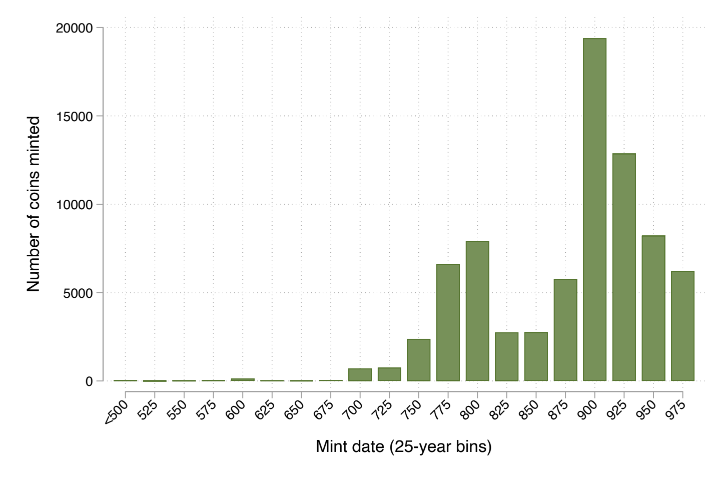
Distribution of coin mint dates



Interval-dated coins use the midpoint of the mint-date interval.

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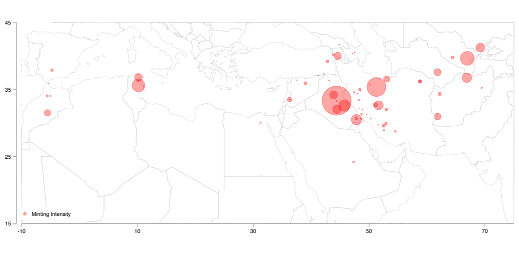
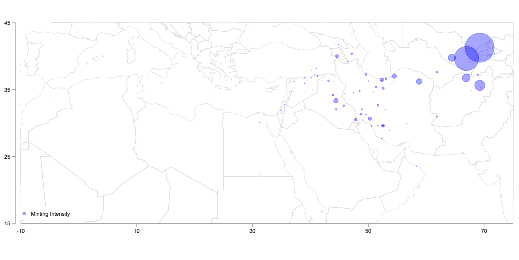
Origins of Islamic silver: before and after 875

Before 875



After 875



Coins with known mint locations and mint years.

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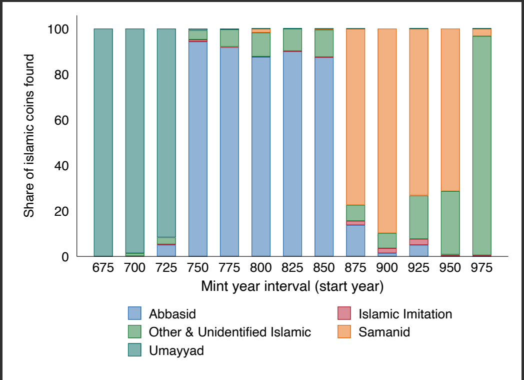
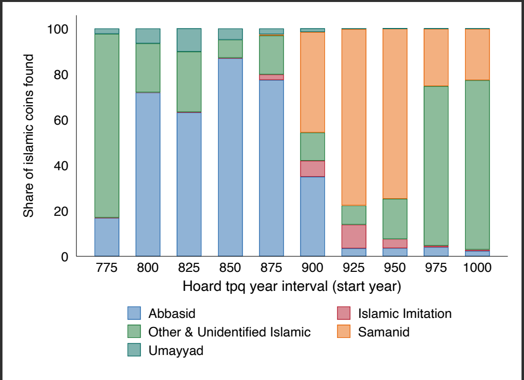
Islamic coins by dynasty

By mint year



By hoard *tpq*



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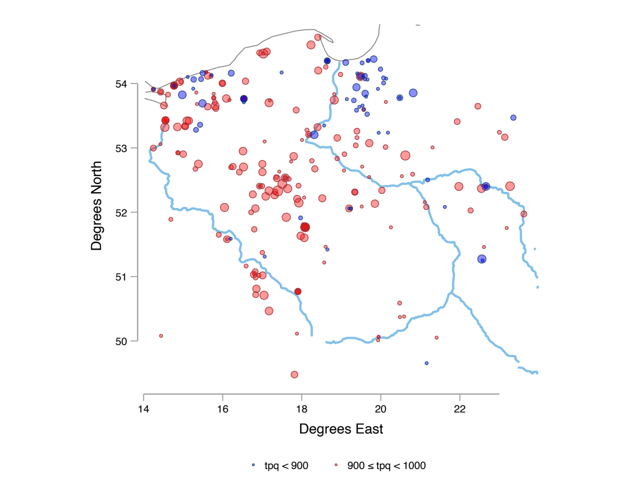
Coin hoards in Poland, 9th–10th century



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Trade disruption IV: controlling for expected exposure

	Dep. var.: $StatePresent_{it}$		Dep. var.: $StatePresent_{h}$		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0324** (0.0099)				
log Number of Coins		0.0895* (0.034)			
Fraction of coins $<$ 20 y.o.			3.576* (1.51)		
Fraction of coins $<$ 40 y.o.				3.333** (1.27)	
Log coin age at deposit					-0.493** (0.12)
Hoard Size Controls			Yes	Yes	
$\mathbb{E}_1[MinTravelTime]_{it}$	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Mint FE					
Unit of Observation	Region $\times$ Time	Region $\times$ Time	Hoard	Hoard	Coin
First stage w. id. robust F	3.812	9.370	5.515	6.817	13.64
R^2	0.134	-1.326	-17.76	-14.11	-12.40
Observations	429	429	916	916	105837

$^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Conflict timing fixed; disruption location varies in the assumed shock process.

▸ Back

Trade disruption IV: alternative expected-exposure control

	Dep. var.: StatePresent_{it}		Dep. var.: StatePresent_{h}		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0331**				
	(0.011)				
log Number of Coins		0.100*			
		(0.042)			
Fraction of coins $<$ 20 y.o.			2.983*		
			(1.33)		
Fraction of coins $<$ 40 y.o.				3.015*	
				(1.31)	
Log coin age at deposit					-0.353**
					(0.11)
Hoard Size Controls			Yes	Yes	
$\mathbb{E}_2[\text{MinTravelTime}]_{it}$	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Mint FE					
Unit of Observation	Region $\times$ Time	Region $\times$ Time	Hoard	Hoard	Coin
First stage w. id. robust F	3.871	6.184	5.035	5.311	9.773
R^2	0.110	-1.775	-12.19	-11.42	-6.133
Observations	429	429	916	916	105837

$^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Disrupted routes fixed; disruption assigned to the 9th or 10th century with equal probability.

▸ Back

Trade disruption IV: including Rus territories

	Dep. var.: StatePresent_{it}		Dep. var.: StatePresent_{h}		
	(1)	(2)	(3)	(4)	(5)
# Hoards	0.0792 (0.047)				
log Number of Coins		0.241* (0.11)			
Fraction of coins $<$ 20 y.o.			2.943* (1.18)		
Fraction of coins $<$ 40 y.o.				2.870** (1.08)	
Log coin age at deposit					-0.598** (0.14)
Hoard Size Controls			Yes	Yes	
Region FE	Yes	Yes	Yes	Yes	Yes
25-year time interval FE	Yes	Yes	Yes	Yes	Yes
Mint FE					
Unit of Observation	Region $\times$ Time	Region $\times$ Time	Hoard	Hoard	Coin
First stage w. id. robust F	3.851	6.277	6.477	7.310	18.32
R^2	-1.831	-4.340	-11.32	-9.767	-13.51
Observations	429	429	916	916	105837

$^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Includes Rus territories. SEs clustered by region in (1)–(2), by coin group in (5).

Signs unchanged; hoard-count coefficient no longer significant.

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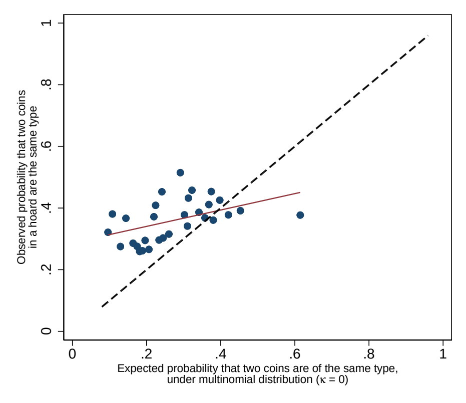
Overdispersion in coin types



- Vertical: probability that two coins from the same hoard have the same type
- Horizontal: independent-draw benchmark, excluding the focal hoard
- Above 45-degree line $\rightarrow$ excess within-hoard similarity

$\hat{\kappa} = 4.99$; independent multinomial limit: $\kappa \rightarrow \infty$.

Back

Polish river traffic estimates

Region	920–940	940–960	960–980	980–1000	Mean 940–980
Lower Vistula	0.77	0.77	0.73	1.05	0.85
Middle Vistula	0.15	0.24	0.10	0.20	0.18
Upper Vistula and Oder	0.40	0.61	0.55	0.74	0.63
Sum all three river regions	1.32	1.62	1.38	2.00	1.67

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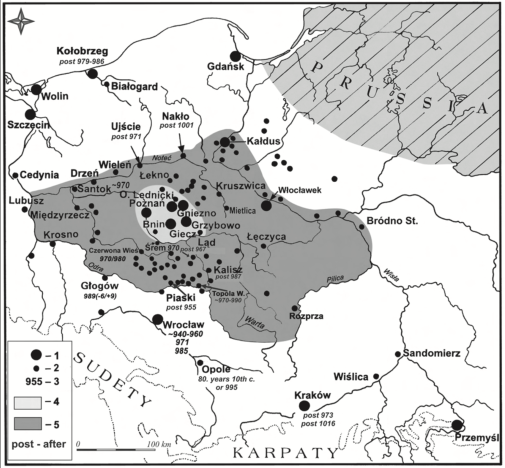
Polish expenditure estimates

Region	920–940	940–960	960–980	980–1000	Mean 940–980
Lower Vistula	0.06	0.08	0.16	0.10	0.11
Middle Vistula	0.01	0.01	0.01	0.00	0.00
Upper Vistula and Oder	0.17	0.16	0.31	0.16	0.21
Sum all three river regions	0.24	0.25	0.48	0.26	0.33

Back

First Polish state under the Piast dynasty: Kara (2020)



Poland, ca. 960. Source: Kara (2020).

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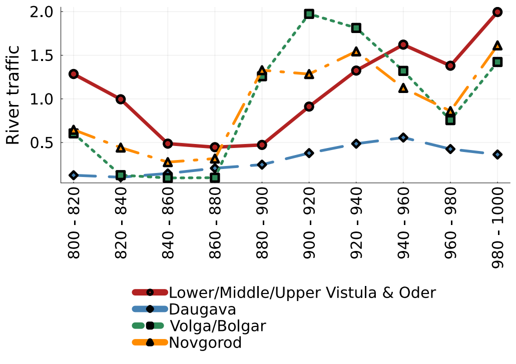
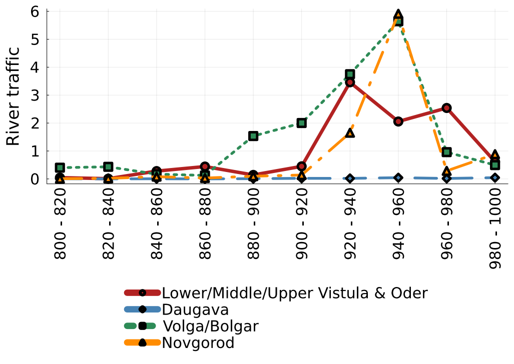
Alternative trade model: within-hoard identification

Baseline



Within-hoard shares + gravity



Alternative avoids identifying coin stocks from hoard counts; reconstructed Polish traffic is qualitatively similar.

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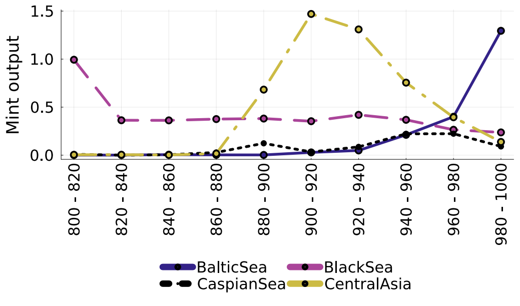
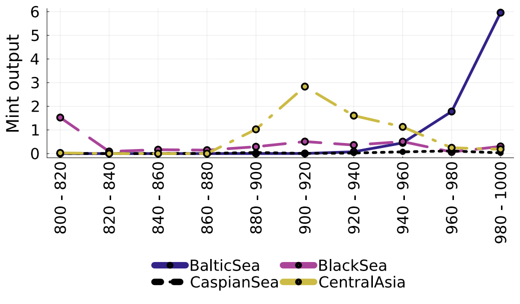
Alternative trade model: estimated mint outputs

Baseline



Within-hoard shares + gravity



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Full parameter estimates

	Meaning	Estimate	95% CI	Interpretation
σ	Spatial decay of production tax value	0.068		Break-even $\approx 1/\sigma \approx 15$ days
ω	Viking cost at the coastline	2.06		$\approx 2.1\times$ a cell's production at the shore
$\kappa_{V,P}$	common Viking and Prague penetration scale	4.62		both costs halve every ≈ 3.2 round-trip travel days
τ_r	toll per unit of (normalized) traffic	5.09		a high-traffic river cell worth ≈ 6.1 cells
λ	frontier defense per edge	1.73		one open frontier edge costs ≈ 1.7 cells
β_R	held-river defensive-side discount	0.94		an eligible side of a held river cell costs $\approx 39\%$ of an open one
ρ	rival-polity cost at Prague	50.00		≈ 50.0 cell-equivalents at Prague

Point estimates; 95% confidence intervals are not reported in the underlying table.

Back

Experimental archaeology: travel speeds

Krampmacken expedition on the Vistula, 1983.

Source: Edberg (2009).

Calibrated travel speeds: 8 km/h downstream; 1 km/h upstream and at portages.

Back

Coin flow model: Summary

Objective: estimate coins "traffic" on rivers. Strategy:

1. Law of motion for total coins in each location:

$$\text{CoinsStock}_{i,t+1} = \underbrace{(1-\lambda)\text{CoinsStock}_{i,t} + \text{NewlyMintedCoins}_{i,t}}_{\text{potentially used for outflows}} + \text{InflowsFromOtherRegions}_{i,t}$$

2. Three assumptions:
   2.1 Coins are fungible ($\rightarrow$ mint date and mint don't matter).
   Then: *flows* of coins of each vintage are proportional to *stocks* of each vintage $\rightarrow$ can be identified from *distribution* of coin types in h
   2.2 Hoards are random samples from the population of coin vintages in each i, t
   2.3 $\mathbb{E}(\text{Q'ty of hoards}_{i,t}) \propto \text{CoinsStock}_{i,t}$

   Then: bilateral coin flows are identified from the data.

3. Assign each bilateral flow to the shortest river path between region centers.

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